

Chapter 4- AC Circuits

Topics to be covered: AC waveform definitions, form factor, peak factor, study of R-L, R-C, RLC series circuit, R-L-C parallel circuit, phasor representation in polar and rectangular form, concept of impedance, admittance, active, reactive, apparent and complex power, power factor, 3 phase Balanced AC Circuits (Y- Δ & Δ -Y).

3.1 Introduction:

An Alternating current (a.c) is the current which changes periodically both in magnitude and direction.

Such change in magnitude and direction is measured in terms of cycles. Each cycle of a.c. consists of two half cycles namely positive cycle and negative cycle. Current increases in magnitude, in one particular direction, attains maximum and starts decreasing, passing through zero it increases in opposite direction and behaves similarly. The Fig. 3.1 (b) shows a graph of alternating current against time.

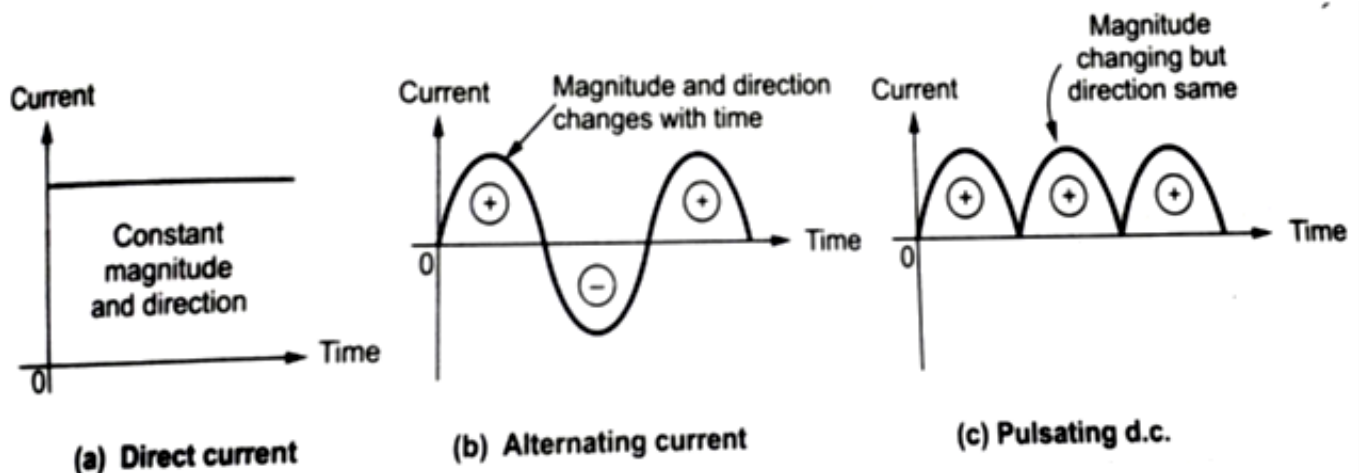


Fig. 3.1

In practice some waveforms are available in which magnitude changes but its direction remains same as positive or negative. This is shown in the Fig. 3.1(c). Such waveform is called pulsating d.c. The waveform obtained as output of full wave rectifier is an example of pulsating d.c.

3.2 Advantages of A.C.

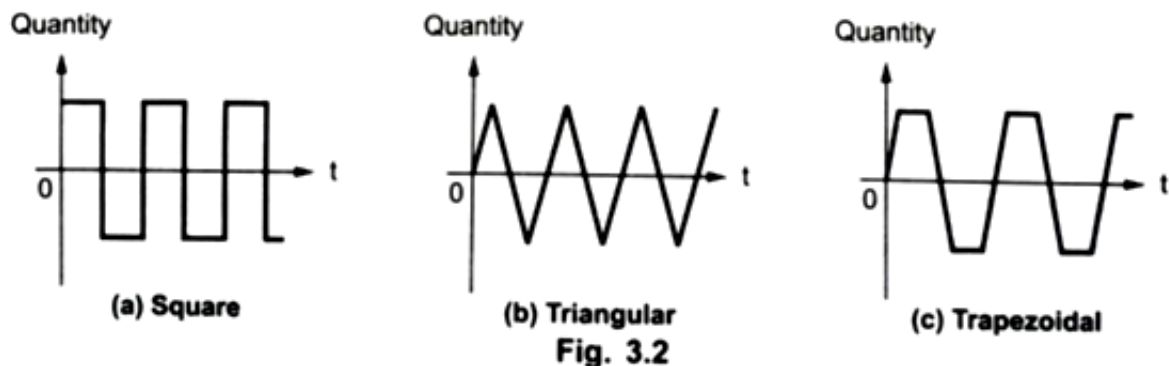
The various advantages of a.c. are,

1. The voltages in a.c. system can be raised or lowered with the help of a device called transformer. In d.c. system, raising and lowering of voltages is not so easy.
2. As the voltages can be raised, electrical transmission at high voltages is possible. Now, higher the voltage, lesser is the current flowing through transmission line. Less the current, lesser are the copper losses and lesser is the **conducting material** required. This makes a.c. transmission always economical and efficient.
3. It is possible to build up high a.c. voltage; high speed a.c. generators of large capacities. The construction and cost of such generators are very low. High a.c. voltages of about 11 kV can be generated and can be raised upto 220 kV for transmission purpose at sending end, while can be lowered down at 400 V at receiving end. This is not possible in case of d.c.
4. A.C. electrical motors are simple in construction, are cheaper and require less attention from maintenance point of view.
5. Whenever it is necessary, a.c. supply can be easily converted to obtain d.c. supply. This is required as d.c. is very much essential for the applications like cranes, printing process, battery charging, telephone system, etc. But, such requirement of d.c. is very small compared to a.c.

Due to these advantages, a.c. is used extensively in practice and hence, it is necessary to study a.c. fundamentals.

3.3 Types of A.C. Waveforms

The waveform of alternating voltage or current is shown purely sinusoidal in the Fig. 3.1 (b). But, in practice, a quantity which undergoes variations in its instantaneous values, in magnitude as well as direction with respect to some zero reference is called an **alternating quantity**. The graph of such quantity against time is called its **waveform**. Various types of alternating waveforms other than sinusoidal are shown in the Fig. 3.2 (a), (b) and (c).



Out of all these types of waveforms, pure sinusoidal waveform is preferred for a.c. systems

3.3.1 Advantages of Purely Sinusoidal Waveform

- 1) Mathematically, it is very easy to write the equations for purely sinusoidal waveform.
- 2) Any other type of waveform can be resolved into a series of sine or cosine waves of fundamental and higher frequencies, sum of all these waves gives the original wave form. Hence, it is always better to have sinusoidal waveform as the standard waveform.
- 3) The sine and cosine waves are the only waves which can pass through linear circuits containing resistance, inductance and capacitance without distortion. In case of other waveforms, there is a possibility of distortion when it passes through a linear circuit.
- 4) The integration and derivative of a sinusoidal function is again a sinusoidal function. This makes the analysis of linear electrical networks with sinusoidal inputs, very easy.

3.4 Generation of A.C. Voltage

The machines which are used to generate electrical voltages are called **generators**. The generators which generate purely sinusoidal a.c. voltages are called **alternators**.

The basic principle of an alternator is the principle of electromagnetic induction. The sine wave is generated according to **Faraday's law of electromagnetic induction**. It says that whenever there is a relative motion between the conductor and the magnetic field in which it is kept, an e.m.f. gets induced in the conductor. The relative motion may exist because of movement of conductors with respect to magnetic field or movement of magnetic field with respect to conductor. Such an induced e.m.f. then can be used to supply the electrical load.

Let us see how an alternator produces a sine wave, with the help of simplest form of an alternator called **single turn** or **single loop alternator**.

3.4.1 Single Turn Alternator

Construction : It consists of a permanent magnet of two poles. A single turn rectangular coil is kept in the vicinity of the permanent magnet. The coil is made up of same conducting material like copper or aluminium. The coil is made up of two conductors namely a-b and c-d. Such two conductors are connected at one end to form a coil. This is shown in the Fig. 3.3.

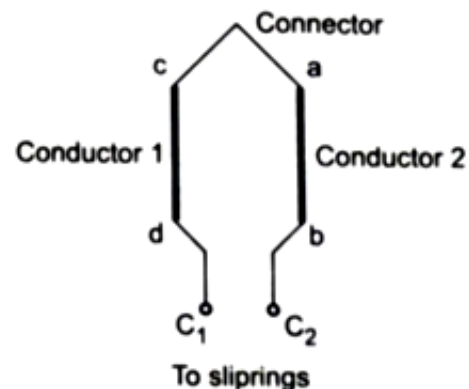


Fig. 3.3 Single turn coil

The coil is so placed that it can be rotated about its own axis in clockwise or anticlockwise direction. The remaining two ends C1 and C2 of the coil are connected to the rings mounted on the shaft called **slip rings**. Slip rings are also rotating members of the alternator. The two brushes P and Q are resting on the slip rings. The **brushes are stationary** and just making contact with the slip rings. The slip rings and brush assembly is **necessary** to collect the current induced in the rotating coil and make it available to the **stationary** external resistance. The overall construction is shown in the Fig. 3.4.

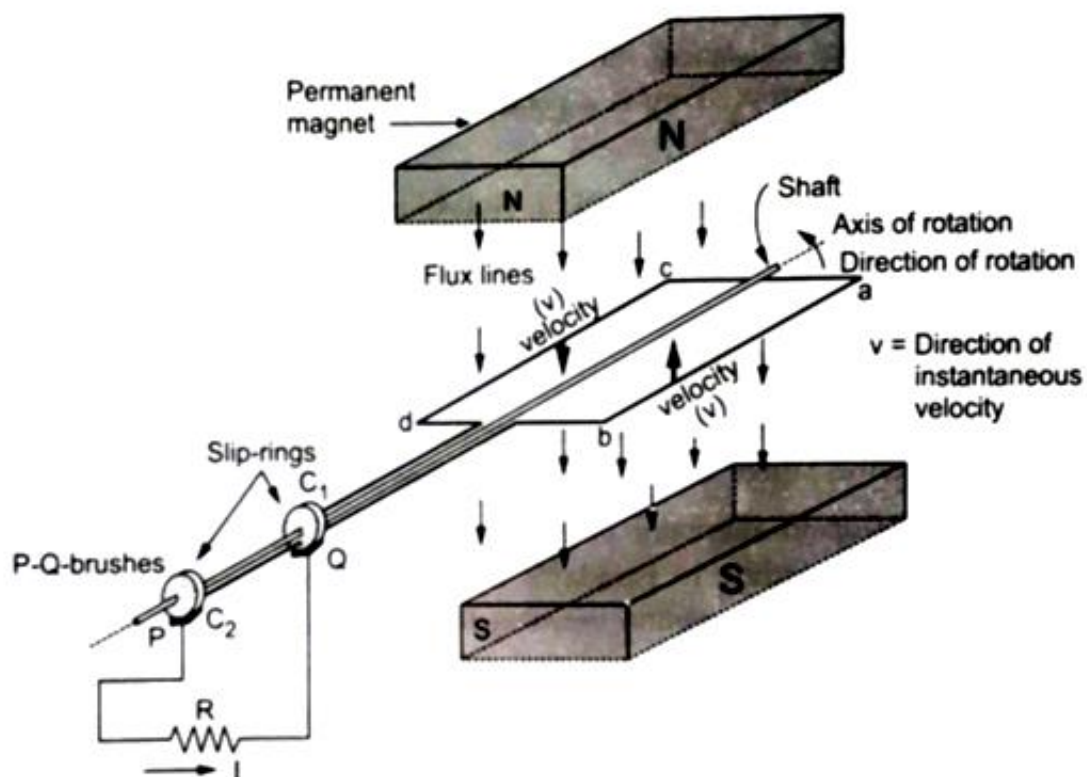


Fig. 3.4 Single turn alternator

Working : The coil is rotated in anticlockwise direction. While rotating, the conductors ab and cd cut the lines of flux of the permanent magnet. Due to Faraday's law of electromagnetic induction, an e.m.f. gets induced in the conductors. This e.m.f. drives a current through resistance R connected across the brushes P and Q. The magnitude of the induced e.m.f. depends on the position of the coil in the magnetic field. Let us see the relation between magnitude of the induced e.m.f. and the positions of the coil. Consider different instants and the different positions of the coil.

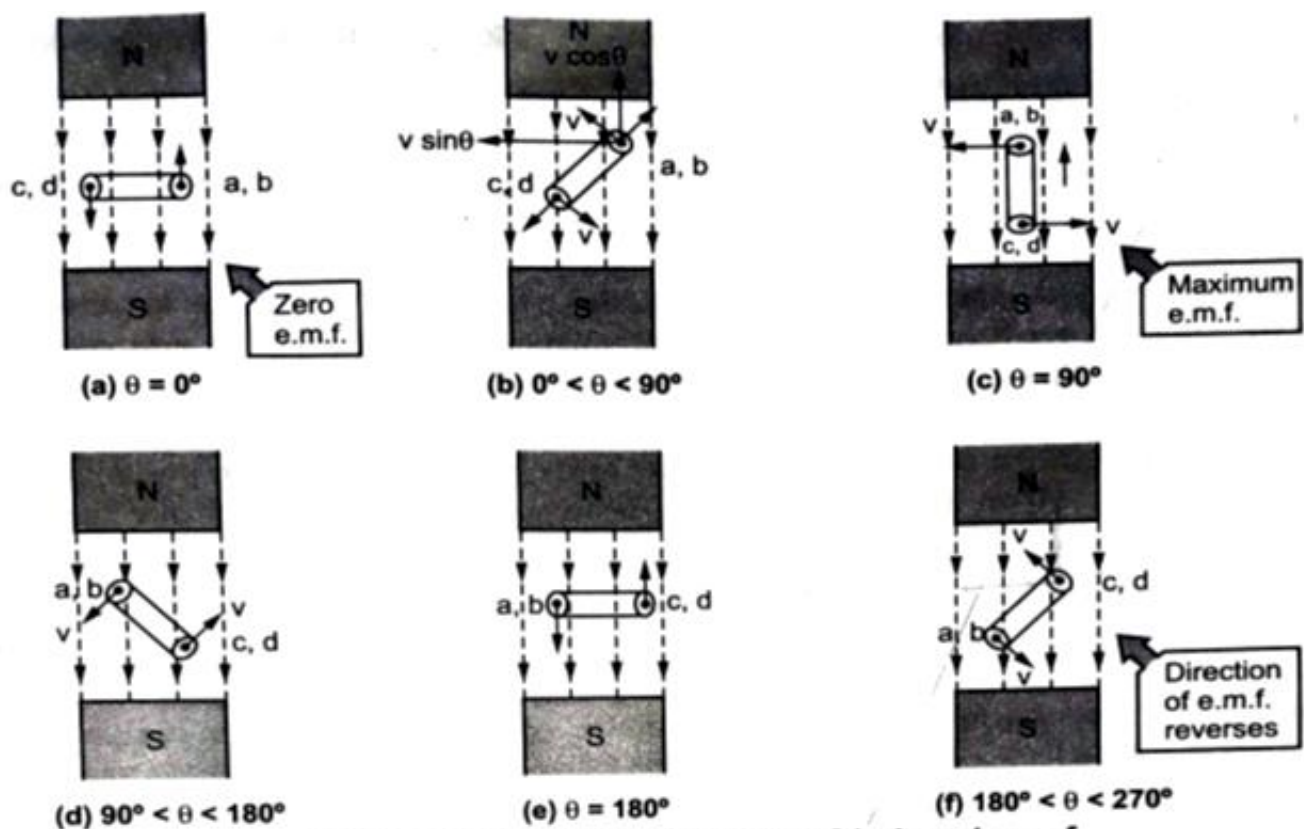


Fig. 3.5 The different instants of induced e.m.f.

The angle θ is measured from plane of the magnetic flux

Instant 1 : Let the initial position of the coil be as shown in the Fig. 3.4. The plane of the coil is perpendicular to the direction of the magnetic field. The instantaneous component of velocity of conductors ab and cd , is parallel to the magnetic field as shown and there cannot be the cutting of the flux lines by the conductors. Hence, no e.m.f. will be generated in the conductors ab and cd and no current will flow through the external resistance R . This position can be represented by considering the front view of the Fig. 3.4 as shown in the Fig. 3.5 (a).

Instant 2 : When the coil is rotated in anticlockwise direction through some angle θ , then the velocity will have two components $v \sin \theta$ perpendicular to flux lines and $v \cos \theta$ parallel to the flux lines. Due to $v \sin \theta$ component, there will be cutting of the flux and proportionally, there will be induced e.m.f. in the conductors ab and cd . This e.m.f. will drive a current through the external resistance R . This is shown in the Fig. 3.5 (b).

Instant 3 : As angle ' θ ' increases, the component of velocity acting perpendicular to flux lines increases, hence induced e.m.f. also increases. At $\theta = 90^\circ$, the plane of the coil is parallel to the plane of the magnetic field while the component of velocity cutting the lines of flux is at its maximum. So, induced e.m.f. in this position, is at its maximum value. This is shown in the Fig. 3.5 (c).

So, as θ increases from 0° to 90° , e.m.f. induced in the conductors increases gradually from 0 to maximum value. The current through external resistance R also varies according to the induced e.m.f.

Instant 4 : As the coil continues to rotate further from $\theta = 90^\circ$ to 180° , the component of velocity, perpendicular to magnetic field starts decreasing, hence, gradually decreasing the magnitude of the induced e.m.f. This is shown in the Fig. 3.5 (d).

Instant 5 : In this position, the velocity component is fully parallel to the lines of flux similar to the instant 1.

Hence, there is no cutting of flux and hence, no induced e.m.f. in both the conductors. Hence, current through external circuit is also zero.

Instant 6 : As the coil rotates beyond $\theta = 180^\circ$, the conductor ab upto now cutting flux lines in one particular direction reverses the direction of cutting the flux lines. Similar is the behaviour of conductor cd. So, direction of induced e.m.f. in conductor ab is opposite to the direction of induced e.m.f. in it for the rotation of $\theta = 0^\circ$ to 180° . Similarly, the direction of induced e.m.f. in conductor cd also reverses. This change in direction of induced e.m.f. occurs because the direction of rotation of conductors ab and cd reverses with respect to the field as θ varies from 180° to 360° . This process continues as coil rotates further. At $\theta = 270^\circ$ again, the induced e.m.f. achieves its maximum value but the direction of this e.m.f. in both the conductors is opposite to the previous maximum position i.e. at $\theta = 90^\circ$. From $\theta = 270^\circ$ to 360° , induced e.m.f. decreases without change in direction and at $\theta = 360^\circ$, coil achieves the starting position with zero induced e.m.f.

So, as θ varies from 0° to 360° , the e.m.f. in a conductor ab or cd varies in an alternating manner i.e. zero, increasing to achieve maximum in one direction, decreasing to zero, increasing to achieve maximum in other direction and again decreasing to zero. This set of variation repeats for every revolution as the conductors rotate in a circular motion with a certain speed.

This variation of e.m.f. in a conductor can be graphically represented.

3.4.2 Graphical Representation of the Induced E.M.F.

The instantaneous values of the induced e.m.f. in any conductor, as it is rotated from $\theta = 0^\circ$ to 360° , i.e. through one complete revolution can be represented as shown in the Fig. 3.6.

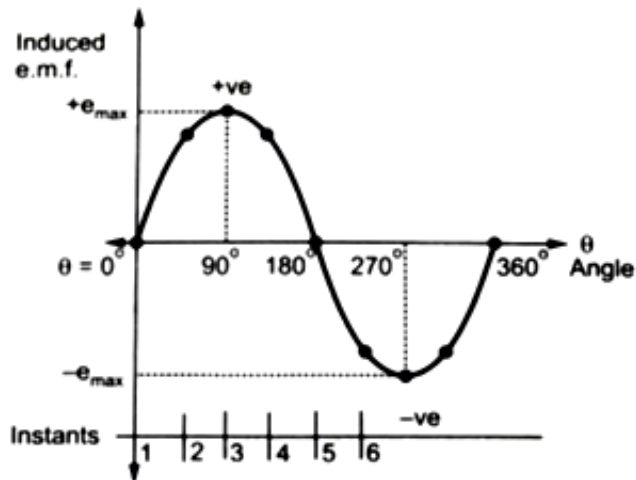
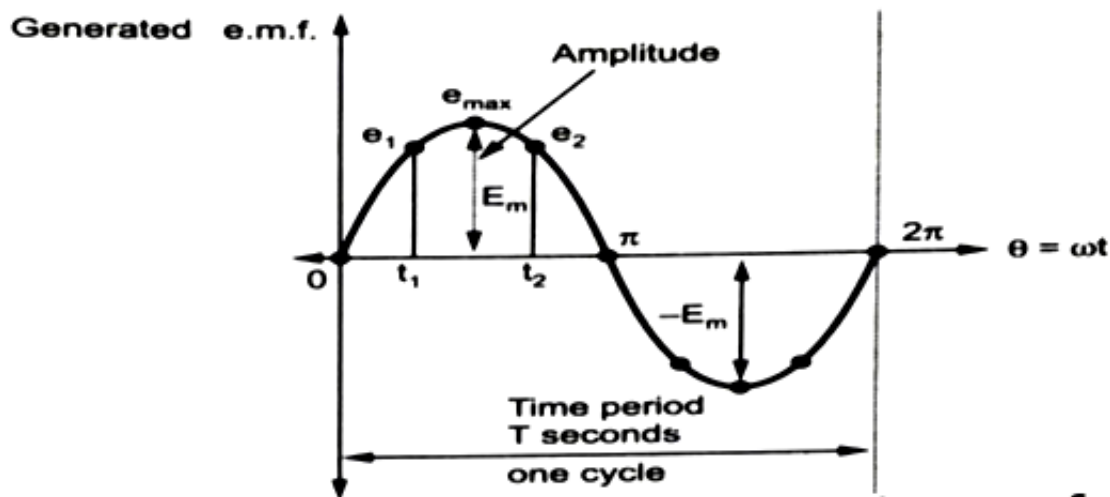


Fig. 3.6 Graphical representation of the induced e.m.f.

From the Fig. 3.6, it is clear that the waveform generated by the instantaneous values of the induced e.m.f. in any conductor (ab or cd) is **purely sinusoidal** in nature.

3.5 Important Definitions



The important terms/definitions which are frequently used while dealing with a.c circuits are

1. Alternating quantity: An alternating quantity is one which acts in alternate positive and negative directions, whose magnitude undergoes a definite series of changes in definite intervals of time and in which the sequence of changes while negative is identical with the sequence of changes while positive.

2. Waveform: “The graph between an alternating quantity (voltage or current) and time is called waveform”, generally, alternating quantity is depicted along the Y-axis and time along the X-axis.

3. Instantaneous value: The value of an alternating quantity at any instant is called instantaneous value. The instantaneous values of alternating voltages and current are represented by ‘e’ and ‘i’ respectively.

4. Alternation: When an alternating quantity goes through one half cycle (complete set of +ve or –ve values) it completes an alternation.

5. Cycle: When an alternating quantity goes a complete set of +ve and –ve values, it is said to have completed one cycle.

6. Periodic Time: The time taken in seconds by an alternating quantity to complete one cycle is known as periodic time and is denoted by ‘T’.

7. Frequency: The number of cycles completed per second by an alternating quantity is known as frequency and is denoted by ‘f’. In the SI system, the frequency is expressed in hertz.

In India, the standard frequency for power supply is 50 Hz. It means that alternating voltage or current completes 50 cycles in one second.

8. Amplitude: The maximum value, positive or negative, which an alternating quantity attains during one complete cycle, is called amplitude or peak value or maximum value. The amplitude of alternating voltage and current is represented by E_m and I_m respectively.

3.6 Equation of an Alternating Quantity

For the derivation of the equation of an alternating quantity, consider single turn, 2 pole alternator discussed earlier. The coil is rotated with constant angular velocity in the magnetic field. An alternating e.m.f. induced is purely sinusoidal in nature.

Let B = Flux density of the magnetic field in Wb/m^2
 l = Active length of each conductor in metres
 r = Radius of circular path traced by conductors in metres.
 ω = Angular velocity of coil in radians/second
 v = Linear velocity of each conductor in m/sec.

Now, $v = r \omega$

Consider an instant where coil has rotated through angle θ from the position corresponding to $\theta = 0^\circ$ i.e. from the instant when induced e.m.f. is zero. It requires time 't' to rotate through θ . So, θ in radians can be expressed as,

$$\theta = \omega t \text{ radians}$$

The position of the coil is shown in the Fig. 3.9 (a). The instantaneous peripheral velocity of any conductor can be resolved into two components as shown in the Fig. 3.9 (b).

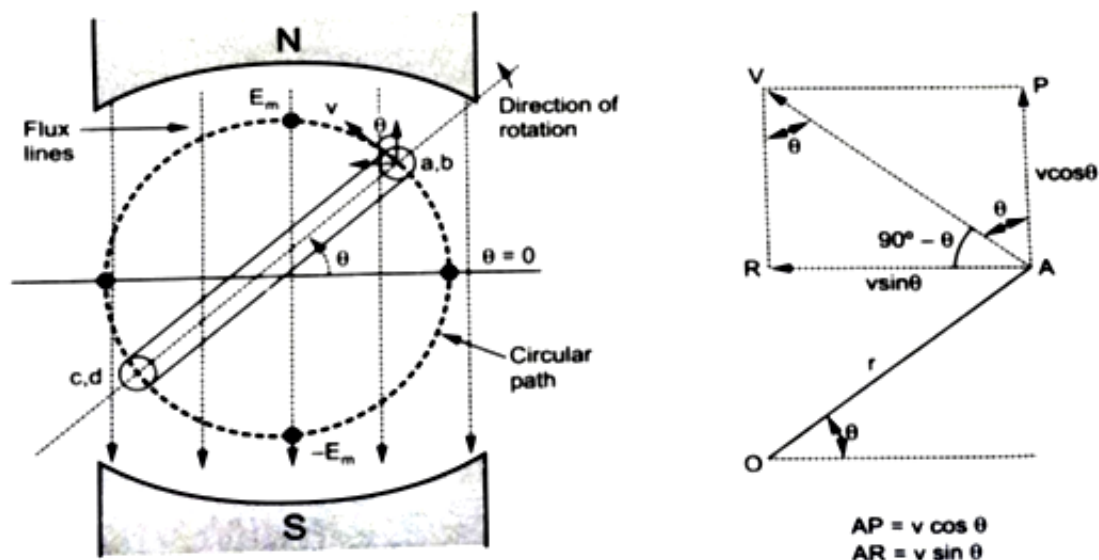


Fig. 3.9 Instantaneous value of an induced e.m.f.

The components of velocity, v are,

- 1) Parallel to the magnetic flux lines, $(AP) = v \cos \theta$
- 2) Perpendicular to the magnetic flux lines, $(AR) = v \sin \theta$

Out of the two, due to the component parallel to the flux, there cannot be the generation of the e.m.f. as there cannot be the cutting of the flux lines. Hence, the component which is acting perpendicular to the magnetic flux lines i.e. $v \sin \theta$ is solely responsible for the generation of the e.m.f.

According to the Faraday's law of electromagnetic induction, the expression for the generated e.m.f. in each conductor is,

$$e = B l v \sin \theta \quad \text{volts}$$

The active length ' l ' means the length of the conductor which is under the influence of the magnetic field.

$$\begin{aligned} \text{Now,} \quad E_m &= B l v \\ &= \text{Maximum value of induced e.m.f. in conductor} \end{aligned}$$

This is achieved at $\theta = 90^\circ$ and is the peak value or amplitude of the sinusoidal induced e.m.f.

Hence, equation giving instantaneous value of the generated e.m.f. can be expressed as,

$$e = E_m \sin \theta \quad \text{volts}$$

3.6.1 Different Forms of E.M.F. Equation

$$\begin{aligned} \text{Now,} \quad \theta &= \omega t \quad \text{radians} \\ \therefore \quad e &= E_m \sin (\omega t) \end{aligned} \quad \dots (1)$$

$$\begin{aligned} \text{But,} \quad \omega &= 2 \pi f \quad \text{rad/sec.} \\ \therefore \quad e &= E_m \sin (2 \pi f t) \end{aligned} \quad \dots (2)$$

$$\begin{aligned} \text{But,} \quad f &= \frac{1}{T} \quad \text{seconds} \\ \therefore \quad e &= E_m \sin \left(\frac{2 \pi}{T} t \right) \end{aligned} \quad \dots (3)$$

Important Note : In all the above equations, the angle θ is expressed in radians. Hence, while calculating the instantaneous value of the e.m.f., it is necessary to calculate the sine of the angle expressed in radians.

Key Point : Mode of the calculator should be converted to radians, to calculate the sine of the angle expressed in radians, before substituting in any of the above equations.

This alternating e.m.f. drives a current through the electrical load which also varies in similar manner.

Its frequency is the same as the frequency of the generated e.m.f. Hence, it can be expressed as,

$$i = I_m \sin \theta$$

where I_m = Maximum or peak value of the current. This maximum value depends on the resistance of the electrical circuit to which an e.m.f. is applied. The instantaneous value of this sinusoidal current set up by the e.m.f. can be expressed as,

$$i = I_m \sin \omega t$$

or

$$i = I_m \sin 2 \pi f t$$

or

$$i = I_m \sin \left(\frac{2 \pi t}{T} \right)$$

Ratings of Alternating Quantity

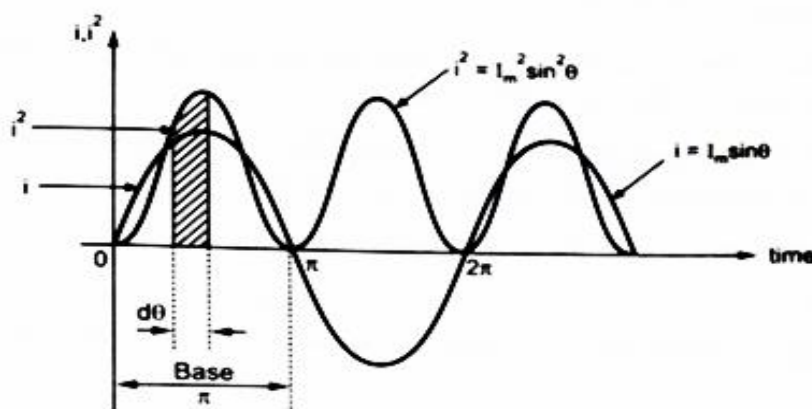
Root-mean-square (R.M.S.) Value or Effective value

The r.m.s. or effective value, of an alternating current is defined as that steady current which when flowing through a given resistance for a given time produces the same amount of heat as produced by the alternating current, when flowing through the same resistance for the same time.

Let us take two circuits with identical resistance, but one is connected to a battery and the other to a sinusoidal voltage source. Wattmeters are employed to measure heat power in each circuit. The voltage applied to each circuit is so adjusted that the heat power produced in each circuit is the same. In this event the direct current 'I' will be equal to $I_m/\sqrt{2}$, which is termed as r.m.s. value of the sinusoidal current.

The following method is used for finding the r.m.s. or effective value of sinusoidal waves.

Consider sinusoidally varying alternating current and square of this current as shown in the Figure



The current $i = I_m \sin \theta$ while

$$\text{Square of current } i^2 = I_m^2 \sin^2 \theta$$

Area of curve over half a cycle can be calculated by considering an interval $d\theta$ as shown.

$$\text{Area of square curve over half cycle} = \int_0^{\pi} i^2 d\theta \text{ and Length of the base is } \pi.$$

$\therefore$ Average value of square of the current over half cycle

$$\begin{aligned} &= \frac{\text{Area of curve over half cycle}}{\text{Length of base over half cycle}} = \frac{\int_0^{\pi} i^2 d\theta}{\pi} \\ &= \frac{1}{\pi} \int_0^{\pi} i^2 d\theta = \frac{1}{\pi} \int_0^{\pi} I_m^2 \sin^2 \theta d\theta = \frac{I_m^2}{\pi} \left[\frac{1 - \cos 2\theta}{2} \right] d\theta \\ &= \frac{I_m^2}{\pi} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\pi} = \frac{I_m^2}{2\pi} [\pi] \\ &= \frac{I_m^2}{2} \end{aligned}$$

Hence, root mean square value i.e. r.m.s. value can be calculated as,

$$\begin{aligned} I_{\text{r.m.s.}} &= \sqrt{\text{Mean or average of square of current}} = \sqrt{\frac{I_m^2}{2}} \\ &= \frac{I_m}{\sqrt{2}} \end{aligned}$$

$$\therefore \boxed{I_{\text{r.m.s.}} = 0.707 I_m}$$

The r.m.s. value of the sinusoidal alternating current is 0.707 times the maximum or peak value or amplitude of that alternating current.

The above result is also applicable to sinusoidal alternating voltages.

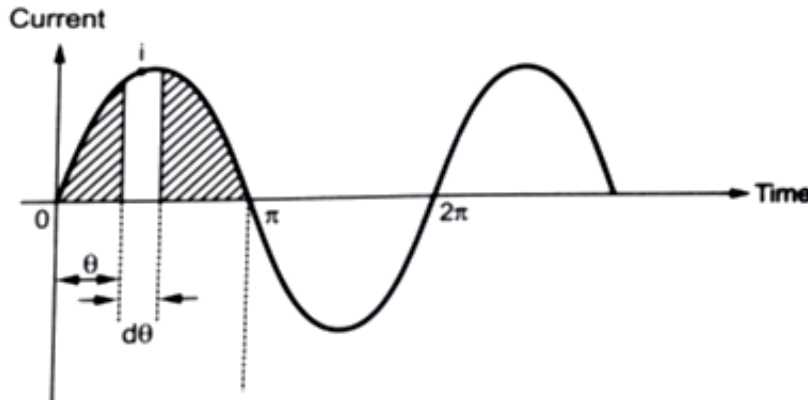
So, we can write,

$$\therefore \boxed{V_{\text{r.m.s.}} = 0.707 V_m}$$

Average Value: The arithmetical average of all the values of an alternating quantity over one cycle is called average value.

In the case of a symmetrical wave e.g. sinusoidal current or voltage wave, the positive half is exactly equal to the negative half, so that the average value over the entire cycle is zero. Hence, in this case, the average value is obtained by adding or integrating the instantaneous values of current over one alternation (half-cycle) only.

Let us take an elementary instant of interval $d\theta$ in the first half-cycle as shown in figure. The average instantaneous values of current in this interval is say 'i' as shown



The average value can be obtained by taking the ratio of area under the curve over half cycle to length of the base for half cycle

$\therefore$

$$I_{av} = \frac{\text{Area under curve for half cycle}}{\text{Length of base over half cycle}}$$

$$= \frac{\int_0^{\pi} i \, d\theta}{\pi}$$

$$= \frac{1}{\pi} \int_0^{\pi} i \, d\theta = \frac{1}{\pi} \int_0^{\pi} I_m \sin \theta \, d\theta = \frac{I_m}{\pi} \int_0^{\pi} \sin \theta$$

$$= \frac{I_m}{\pi} [-\cos \theta]_0^{\pi} = \frac{I_m}{\pi} [-\cos \pi + \cos 0]$$

$$= \frac{I_m}{\pi} [2] = \frac{2 I_m}{\pi}$$

$\therefore$

$$I_{av} = \frac{2}{\pi} I_m$$

i.e.

$$I_{av} = 0.637 I_m$$

Similarly,

$$V_{av} = 0.637 V_m$$

Form Factor: The ratio of effective value (or r.m.s. value) to average value of an alternating quantity (voltage or current) is called form factor, i.e.

$$\text{Form Factor, } K_f = \frac{\text{r ms value of an alternating quantity}}{\text{Average value of an alternating quantity}}$$

For sinusoidal alternating current,

$$K_f = \frac{0.707 I_m}{0.637 I_m} = 1.11$$

For sinusoidal alternating voltage,

$$K_f = \frac{0.707 E_m}{0.637 E_m} = 1.11$$

Hence, the R.M.S. value (of current or voltage) is 1.11 times its average value.

Crest or Peak or Amplitude Factor (Ka): It is defined as the ratio of maximum value to the effective value (r.m.s. value) of an alternating quantity. i.e.,

$$\text{Peak factor, } K_a = \frac{\text{Maximum value of an alternating quantity}}{\text{r ms value of an alternating quantity}}$$

For sinusoidal alternating current,

$$K_a = \frac{I_m}{I_m/\sqrt{2}} = \sqrt{2} = 1.414$$

For sinusoidal alternating voltage,

$$K_a = \frac{E_m}{E_m/\sqrt{2}} = \sqrt{2} = 1.414$$

The knowledge of Crest Factor is particularly important in the testing of dielectric strength of insulating materials; this is because the breakdown of insulating materials depends upon the maximum value of voltage.

Phasor Representation:

An alternating quantity can be represented using

- i) Waveform
- ii) Equations
- iii) Phasor
- iv) Complex form

Complex number

Complex numbers represent in two ways:

1. **Rectangular form:** $C = x + j y$
Where x : is called the real part of C
 y : is called its imaginary part
 $j = -1$

2. **Polar form:** $C = r \angle \theta$
Where ' r ' is the magnitude of C .

Conversion between Rectangular and Polar Forms

1. **To convert from Rectangular form to Polar form**

$$C = x + j y \rightarrow C = r \angle \theta$$

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \tan^{-1}(y/x)$$

2. **To convert from Polar form to Rectangular form:**

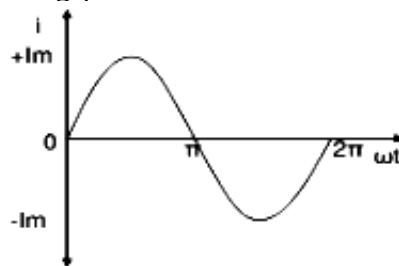
$$C = r \angle \theta \rightarrow C = x + j y$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

Phasor:

A sinusoidal alternating quantity can be represented by a rotating line called a **Phasor**. A Phasor is a line of definite length rotating in anticlockwise direction at a constant angular velocity. The waveform and equation representation of an alternating current is as shown. This sinusoidal quantity can also be represented using phasors.

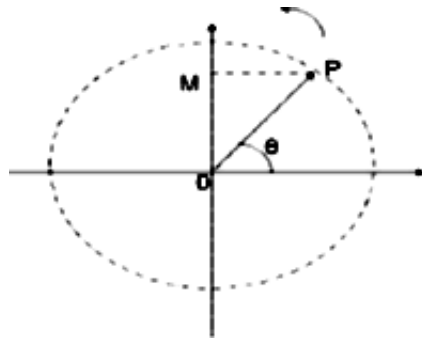


$$i = I_m \sin \omega t$$

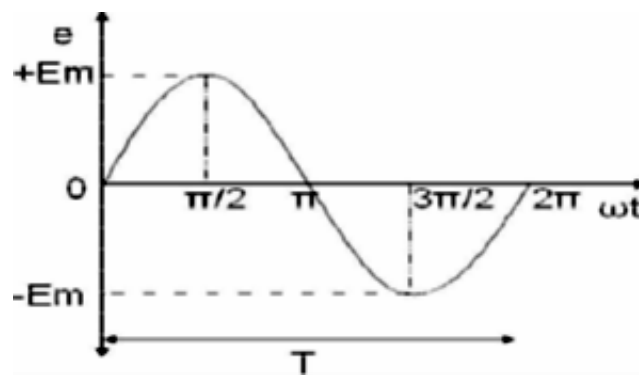
In phasor form the above wave is written as $I_m \angle 0^\circ$

Draw a line OP of length equal to I_m . This line OP rotates in the anticlockwise direction with a uniform angular velocity ω rad/sec and follows the circular trajectory shown in figure. At

any instant, the projection of OP on the y-axis is given by $OM = OP \sin \theta = I_m \sin \omega t$. Hence the line OP is the phasor representation of the sinusoidal current.



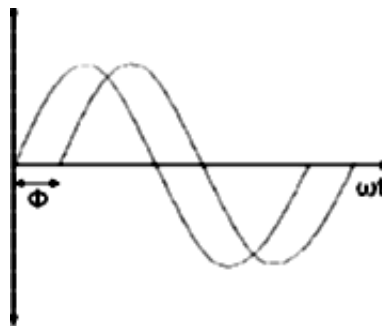
Phase: Phase is defined as the fractional part of time period or cycle through which the quantity has advanced from the selected zero position of reference



Phase of $+E_m$ is $\pi/2$ rad or $T/4$ sec

Phase of $-E_m$ is $3\pi/2$ rad or $3T/4$ sec

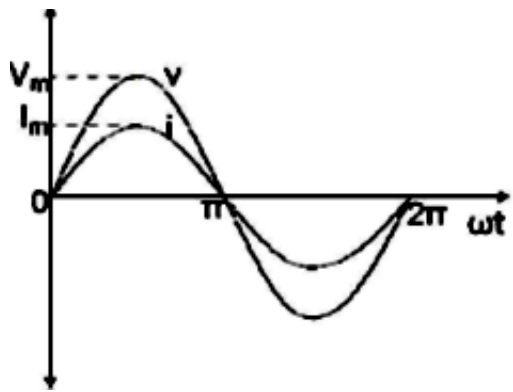
Phase Difference: When two alternating quantities of the same frequency have different zero points, they are said to have a phase difference. The angle between the zero points is the angle of phase difference.



In Phase

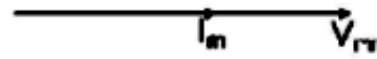
Two waveforms are said to be in phase, when the phase difference between them is zero.

That is the zero points of both the waveforms are same. The waveform, phasor and equation representation of two sinusoidal quantities which are in phase is as shown. The figure shows that the voltage and current are in phase.



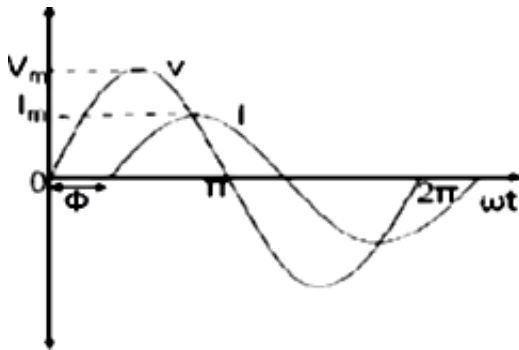
$$v = V_m \sin \omega t$$

$$i = I_m \sin \omega t$$



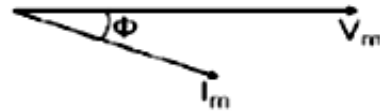
Lagging

In the figure shown, the zero point of the current waveform is after the zero point of the voltage waveform. Hence the current is lagging behind the voltage. The waveform, phasor and equation representation is as shown.



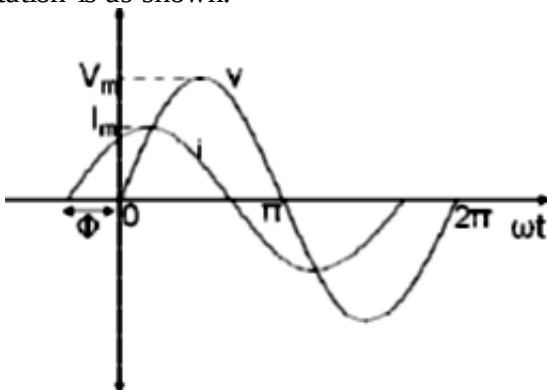
$$v = V_m \sin \omega t$$

$$i = I_m \sin (\omega t - \Phi)$$



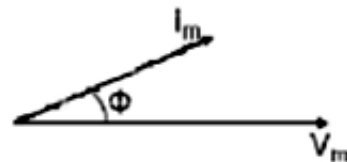
Leading

In the figure shown, the zero point of the current waveform is before the zero point of the voltage waveform. Hence the current is leading the voltage. The waveform, phasor and equation representation is as shown.



$$v = V_m \sin \omega t$$

$$i = I_m \sin (\omega t + \Phi)$$



Steady State Analysis of A.C. circuits

The path for the flow of alternating current is called an a.c. circuit. In a d.c. circuit, the current flowing through the circuit is given by the simple relation $I = V/R$. However, in an a.c. circuit, voltage and current change from instant to instant and so give rise to magnetic (inductive) and electrostatic (capacitive) effects. So, in an a.c. circuit, inductance and capacitance must be considered in addition to resistance.

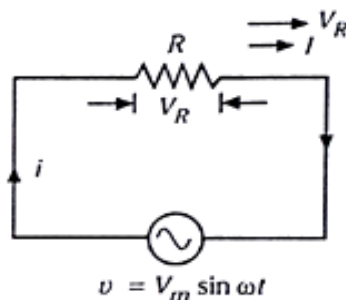
We shall now deal with the following a.c. circuits:

1. AC circuit containing pure ohmic resistance only.
2. AC circuit containing pure inductance only.
3. AC circuit containing pure capacitance only.

AC through pure resistive circuit

When an alternating voltage is applied to a pure resistive circuit, electrons flow in one direction during first-half cycle and in opposite direction during second-half cycle, thus constituting alternating current in the circuit.

Let us consider an a.c. circuit with pure resistance R , as shown in figure



Let the applied voltage be given by the equation

$$v = V_m \sin \theta = V_m \sin \omega t \quad \text{---(i)}$$

As a result of this alternating voltage, alternating current ' i ' will flow through the circuit.

The applied voltage has to supply the drop in the resistance, *i.e.*,

$$v = iR$$

Substituting the value of ' v ' from eqn.(i), we get

$$V_m \sin \omega t = iR$$

$$\text{or } i = \frac{V_m}{R} \sin \omega t \quad \text{---(ii)}$$

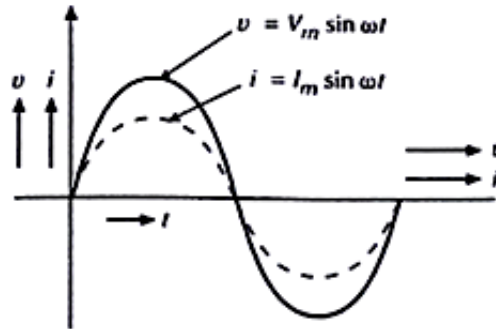
The value of the alternating current 'i' is maximum when $\sin \omega t = 1$,

$$\text{i.e., } I_m = \frac{V_m}{R}$$

$\therefore$ Eqn.(ii) becomes,

$$i = I_m \sin \omega t \quad \dots(\text{iii})$$

From equations (i) and (iii) it is apparent that voltage and current are in phase with each other and it is represented in waveform as shown below.



Power in a pure resistive circuit

The voltage and current are changing at every instant

$$\begin{aligned} \therefore \text{ Instantaneous power, } p &= V_m \sin \omega t \cdot I_m \sin \omega t \\ &= V_m I_m \sin^2 \omega t \\ &= V_m I_m \frac{(1 - \cos 2\omega t)}{2} \\ &= \frac{V_m I_m}{2} - \frac{V_m I_m}{2} \cos 2\omega t \end{aligned}$$

Thus instantaneous power consists of a constant part $\frac{V_m I_m}{2}$ and a fluctuating part $\frac{V_m I_m}{2} \cos 2\omega t$ of frequency double that of voltage and current waves.

The average value of $\frac{V_m I_m}{2} \cos 2\omega t$ over a complete cycle is zero.

So, power for the complete cycle is

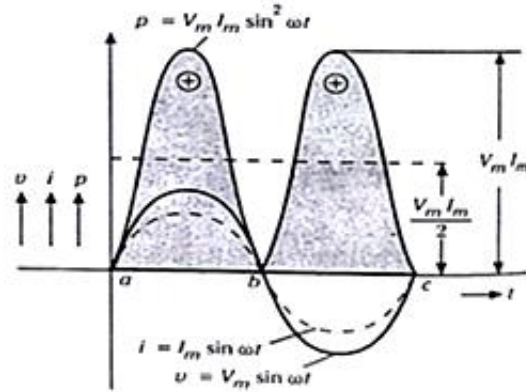
$$P = \frac{V_m I_m}{2} = \frac{V_m}{\sqrt{2}} \times \frac{I_m}{\sqrt{2}}$$

or $P = VI$ watts

where V = r.m.s. value of applied voltage

I = r.m.s. value of the current

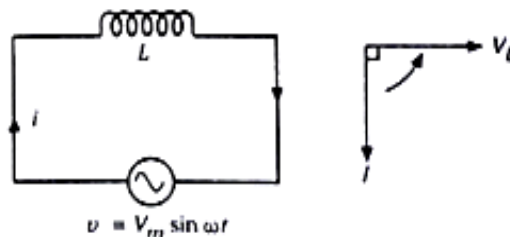
Power curve: The power curve for a purely resistive circuit is as shown in figure below



From the figure it is apparent that power in such a circuit is zero only at the instants a, b and c, when both the voltage and current are zero, it is positive at all other instants. In other words, power is never negative, so that power is always lost in a resistive ac circuit. This power is dissipated as heat.

AC through pure inductive circuit

An inductive coil is a coil with or without an iron core and has negligible resistance. In practice, pure inductance can never be found as inductive coil always has a small resistance. However for the study purpose we will consider a purely inductive coil with zero resistance excited by a ac source as shown in the figure below



On the application of an alternating voltage to the circuit containing a pure inductance, a back e.m.f is produced due to the self inductance of the coil. This back e.m.f. opposes the rise and fall of the current at every stage. The applied voltage has to overcome this self induced e.m.f.

Let the applied voltage be $v = V_m \sin \omega t$

The self-inductance of the coil = L Henry

Self induced e.m.f in the coil is $= e_L = -L \frac{di}{dt}$

Since the applied voltage is equal and opposite to the self induced e.m.f at every instant we have

$$V = -e_L$$

$$\therefore V_m \sin \omega t = -\left(-L \frac{di}{dt}\right)$$

$$\text{or } di = \frac{V_m}{L} \sin \omega t dt$$

Integrating both sides, we get

$$i = \frac{V_m}{L} \int \sin \omega t dt$$

$$\text{or } i = \frac{V_m}{\omega L} (-\cos \omega t) + A$$

Where A is a constant of integration and found to be zero from initial condition

$$\text{So, } i = \frac{-V_m}{\omega L} \cos \omega t$$

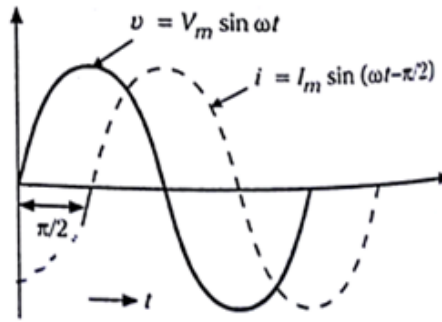
$$\text{or } i = \frac{V_m}{\omega L} \sin \left(\omega t - \frac{\pi}{2} \right)$$

Current will be maximum when $\sin \left(\omega t - \frac{\pi}{2} \right) = 1$, hence, the value of maximum current,

$$I_m = \frac{V_m}{\omega L},$$

$$\text{and } i = I_m \sin \left(\omega t - \frac{\pi}{2} \right).$$

From the expressions for the instantaneous voltage and instantaneous current flowing through a purely inductive coil, it is clear the the current lags the voltage by $\pi/2$ as shown in the waveform below



Inductive Reactance: ωL in the expression $I_m = V_m / \omega L$ is known as inductive reactance and is denoted by X_L , i.e., $X_L = \omega L$. If L is in Henry, ω is in radians per second, then X_L is in Ohms. So, inductive reactance plays a role of resistance.

Power: Instantaneous power is given by

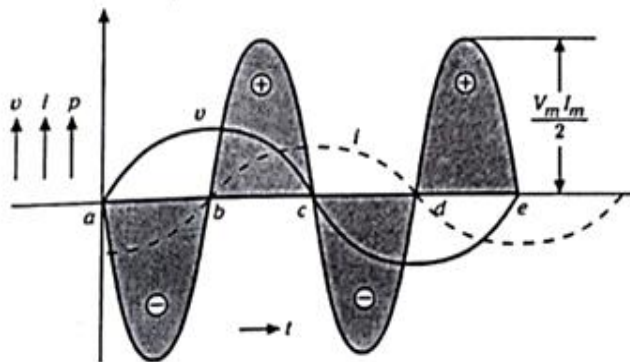
$$\begin{aligned} p &= v \times i = V_m \sin \omega t \cdot I_m \sin \left(\omega t - \frac{\pi}{2} \right) \\ &= -V_m I_m \sin \omega t \cos \omega t \\ &= \frac{-V_m I_m}{2} \sin 2\omega t \end{aligned}$$

The power measured by a wattmeter is the average value of 'p' which is zero, since average value of a sinusoidal quantity of double frequency over a complete cycle is zero. In mathematical form it can be expressed as

$$\text{Power for the whole cycle, } p = -\frac{V_m I_m}{2} \int_0^{2\pi} \sin 2\omega t \, dt = 0$$

Hence the average power absorbed by a pure inductive circuit is zero.

Power Curve: The power curve for a pure inductive circuit is as shown in figure below.



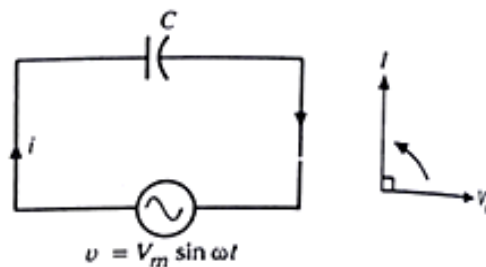
This indicates that power absorbed by the circuit is zero. At the instants a, c and e, voltage is zero so the power is zero; at instants b and d also the power is zero as the current is zero. Between a

and b voltage and current are in opposite directions, so that power is negative and energy is taken from the circuit. Between b and c voltage and current are in same directions, so that power is positive and energy is put back to the circuit. Similarly, between c and d, power is taken from the circuit and between d and e it is put back into the circuit. Hence net power over a complete cycle is zero.

AC through pure capacitive circuit

When an alternating voltage is applied across the plates of the capacitor, it charges in one direction during first-half cycle and then charges in opposite direction as the voltage reverses

Let alternating voltage $v = V_m \sin \omega t$ be applied across a capacitor of capacitance C Farad as shown in the figure



Instantaneous charge, $q = Cv = CV_m \sin \omega t$

Capacitor current is equal to the rate of change of charge, or

$$i = \frac{dq}{dt} = \frac{d}{dt} (CV_m \sin \omega t)$$

$$= \omega CV_m \cos \omega t$$

$$\text{or } i = \frac{V_m}{\frac{1}{\omega C}} \cos \omega t$$

$$\text{or } i = \frac{V_m}{\frac{1}{\omega C}} \sin \left(\omega t + \frac{\pi}{2} \right)$$

The current is maximum when $t = 0$

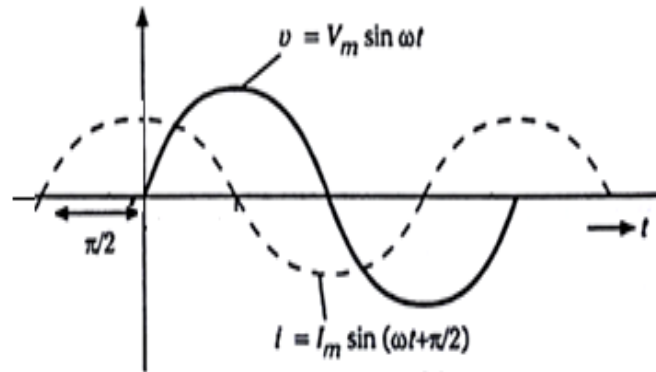
$$\therefore I_m = \frac{V_m}{\frac{1}{\omega C}}$$

Substituting $\frac{V_m}{\frac{1}{\omega C}} = I_m$ in the above expression for instantaneous current, we get

$$i = I_m \sin \left(\omega t + \frac{\pi}{2} \right)$$

Capacitive Reactance: $1/\omega C$ in the expression $I_m = V_m / (1/\omega C)$ is known as capacitive reactance and is denoted by X_C , i.e., $X_C = 1/\omega C$. If C is in Farad, ω is in radians per second, then X_C is in Ohms. So, capacitive reactance plays a role of resistance.

From the expressions for the instantaneous voltage and instantaneous current flowing through a purely capacitive circuit, it is clear that the current leads the voltage by $\pi/2$ as shown in the waveform below



Power : Instantaneous Power,

$$\begin{aligned} p &= vi \\ &= V_m \sin \omega t \cdot I_m \sin \left(\omega t + \frac{\pi}{2} \right) \\ &= V_m I_m \sin \omega t \cos \omega t \\ &= \frac{1}{2} V_m I_m \sin 2\omega t \end{aligned}$$

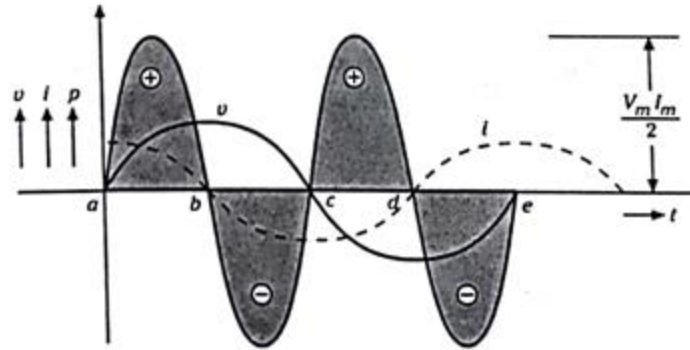
The power measured by a wattmeter is the average value of 'p' which is zero, since average value of a sinusoidal quantity of double frequency over a complete cycle is zero. In mathematical form it can be expressed as

Power for the complete cycle

$$= \frac{1}{2} V_m I_m \int_0^{2\pi} \sin 2\omega t \, dt = 0$$

Hence the average power consumed by a pure capacitive circuit is zero

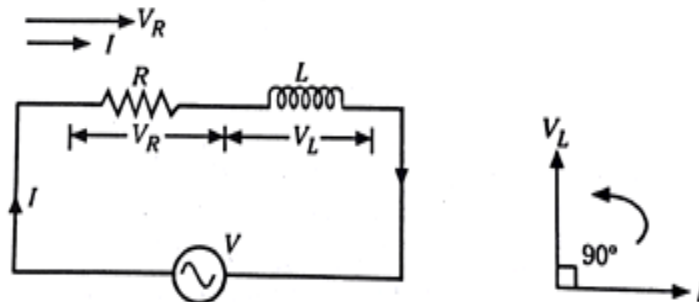
Power Curve: The power curve for a pure inductive circuit is as shown in figure below.



This indicates that power absorbed by the circuit is zero. At the instants a, c and e, voltage is zero so the power is zero; at instants b and d also the power is zero as the current is zero. Between a and b voltage and current are in same directions, so that power is positive and energy is put back to the circuit. Between b and c voltage and current are in opposite directions, so that power is negative and energy is taken from the circuit. Similarly, between c and d, power is put back to the circuit and between d and e it is taken from the circuit. Hence net power over a complete cycle is zero.

AC through series RL circuit:

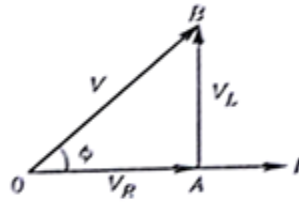
Let us consider an a.c circuit containing a pure resistance R and pure inductance L , as shown in the figure



Let V be the r.m.s value of applied voltage
 I be the r.m.s value of current

Let the Voltage drop across R , $V_R = IR$ (in-phase with I)
 Voltage drop across L , $V_L = IX_L$ (Leading I by 90°)

The voltage drops across these two circuit components are as shown in figure



Voltage vector

From figure vector OA indicates V_R and AB indicate V_L . The applied voltage V is the vector sum of the two i.e. OB

OB is given by

$$OB^2 = OA^2 + AB^2$$

$$\therefore V = \sqrt{V_R^2 + V_L^2} = \sqrt{(IR)^2 + (IX_L)^2}$$

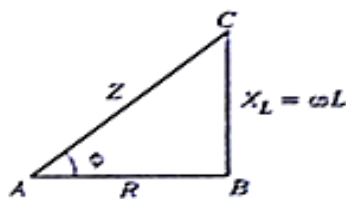
$$= I \sqrt{R^2 + X_L^2}$$

$$\therefore I = \frac{V}{\sqrt{R^2 + X_L^2}}$$

The term $\sqrt{R^2 + X_L^2}$ offers opposition to current flow and is called the impedance (Z) of the circuit. It is measured in ohms.

$$\therefore I = \frac{V}{Z}$$

Referring to the impedance triangle ABC



Impedance triangle

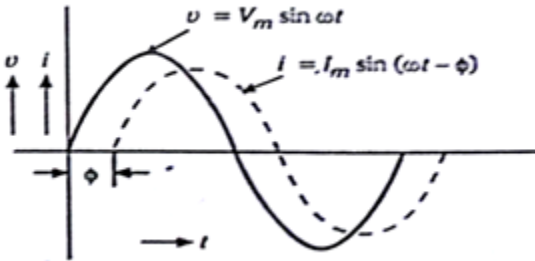
$$Z^2 = R^2 + X_L^2$$

Referring to the voltage vector we observe that the applied voltage V leads the current I by an angle ϕ

$$\tan \phi = \frac{V_L}{V_R} = \frac{I \cdot X_L}{I \cdot R} = \frac{X_L}{R} = \frac{\omega L}{R} = \frac{\text{reactance}}{\text{resistance}}$$

$$\therefore \phi = \tan^{-1} \frac{X_L}{R}$$

This is as shown in the waveforms below



Power :

$$\begin{aligned} \text{Instantaneous power, } p &= vi = V_m \sin \omega t \times I_m \sin (\omega t - \phi) \\ &= V_m I_m \sin \omega t \sin (\omega t - \phi) \\ &= \frac{1}{2} V_m I_m [\cos \phi - \cos (2\omega t - \phi)] \end{aligned}$$

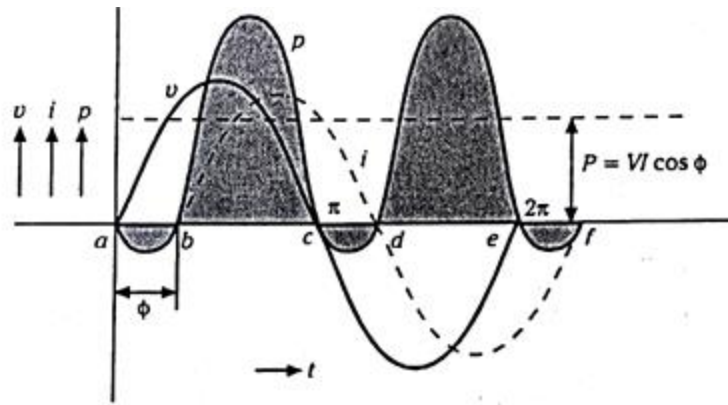
This consists of two parts

- i) Constant part $\frac{1}{2} V_m I_m \cos \phi$ which contributes to real power.
- ii) Sinusoidally varying part $\frac{1}{2} V_m I_m \cos (2\omega t - \phi)$, whose frequency is twice that of the voltage and the current, and whose average value over a complete cycle is zero (so it does not contribute to any power).

$$\begin{aligned} \text{So, average power consumed, } P &= \frac{1}{2} V_m I_m \cos \phi \\ &= \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}} \cos \phi \\ &= VI \cos \phi \end{aligned}$$

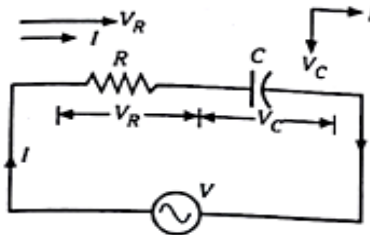
Where, V and I are the r.m.s values

Power consumed can be represented in waveform as shown below.



AC through series RC circuit:

Let us consider an a.c circuit containing a pure resistance R and pure capacitance C , as shown in the figure



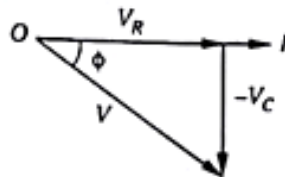
Let, V be the r.m.s value of applied voltage

I be the r.m.s value of current

Let the, Voltage drop across R , $V_R = IR$ (in-phase with I)

Voltage drop across C , $V_C = IX_C$ (Lagging I by 90°)

The voltage drops across these two circuit components are as shown in figure

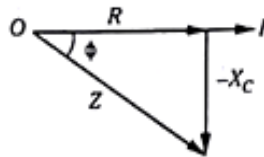


The capacitive reactance is negative, so V_C is in the negative direction of Y-axis, as shown in figure

We have $V = \sqrt{V_R^2 + (-V_C)^2} = \sqrt{(IR)^2 + (-IX_C)^2}$
 $= I \sqrt{R^2 + X_C^2}$
 or $I = \frac{V}{\sqrt{R^2 + X_C^2}} = \frac{V}{Z}$

The denominator, Z is the *impedance* of the circuit, i.e., $Z = \sqrt{R^2 + X_C^2}$.

Referring to the impedance triangle ABC

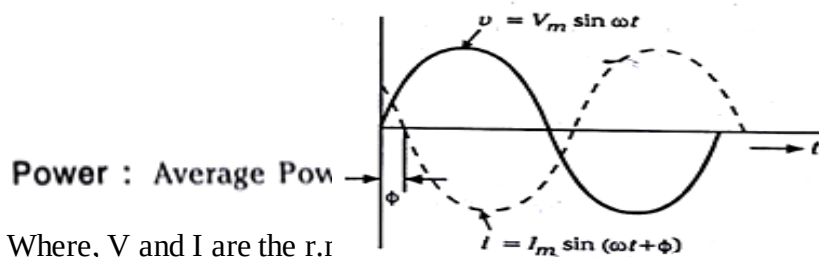


$$\tan \phi = -\frac{X_C}{R}.$$

Power factor, $\cos \phi = \frac{R}{Z}$

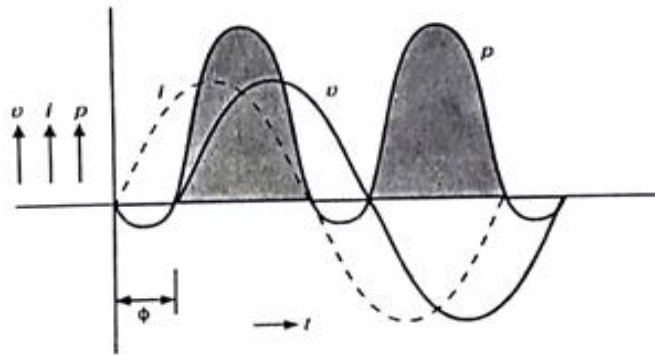
Referring to the voltage vector we observe that the applied voltage V lags the current I by an angle ϕ

This is as shown in the waveforms below



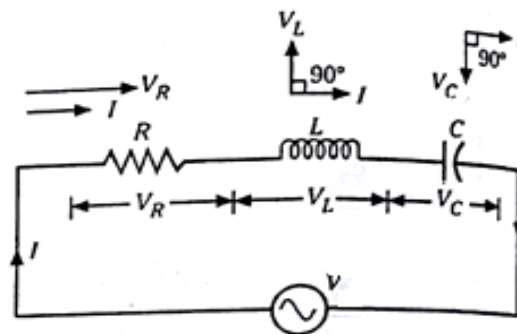
Where, V and I are the r.m.s

Power consumed can be represented in waveform as shown below.



AC through series RLC circuit

Let us consider an a.c circuit containing a pure resistance R , pure inductance L and pure capacitance C , in series with each other across a.c supply as shown in the figure



The ac supply is given by, $v = V_m \sin \omega t$. The circuit draws a current I , there are voltage drops across R , L and C which are given by

- Voltage drop across R , $V_R = I R$ (in-phase with I)
- Voltage drop across L , $V_L = I X_L$ (V_L Leading I by 90°)
- Voltage drop across C , $V_C = I X_C$ (V_C Lagging I by 90°)

The values of I , V_R , V_L , V_C are r.m.s values

According to Kirchhoff's laws, we can write,

$$\bar{V} = \bar{V}_R + \bar{V}_L + \bar{V}_C \quad \dots \text{Phasor addition}$$

Let us see the phasor diagram. Current I is taken as reference as it is common to all the elements.

Following are the steps to draw the phasor diagram :

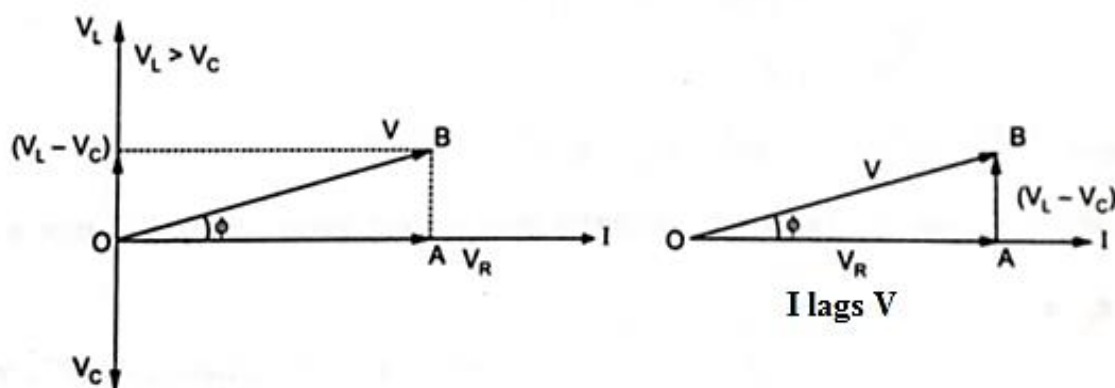
- 1) Take current as reference.
- 2) V_R is in phase with I .
- 3) V_L leads current I by 90° .
- 4) V_C lags current I by 90° .
- 5) Obtain the resultant of V_L and V_C . Both V_L and V_C are in phase opposition (180° out of phase).
- 6) Add that with V_R by law of parallelogram to get the supply voltage.

The phasor diagram depends on the conditions of the magnitudes of V_L and V_C which ultimately depends on the values of X_L and X_C . Let us consider the different cases.

$X_L > X_C$

When $X_L > X_C$, obviously, $I X_L$ i.e. V_L is greater than $I X_C$ i.e. V_C . So, resultant of V_L and V_C will be directed towards V_L i.e. leading current I . Current I will lag the resultant of V_L and V_C i.e. $(V_L - V_C)$.

The circuit is said to be **inductive** in nature. The phasor sum of V_R and $(V_L - V_C)$ gives the resultant supply voltage, V . This is shown in the Fig.



From the voltage triangle, $V = \sqrt{(V_R)^2 + (V_L - V_C)^2} = \sqrt{(IR)^2 + (IX_L - IX_C)^2}$

$$= I \sqrt{(R)^2 + (X_L - X_C)^2}$$

$$\therefore V = I Z$$

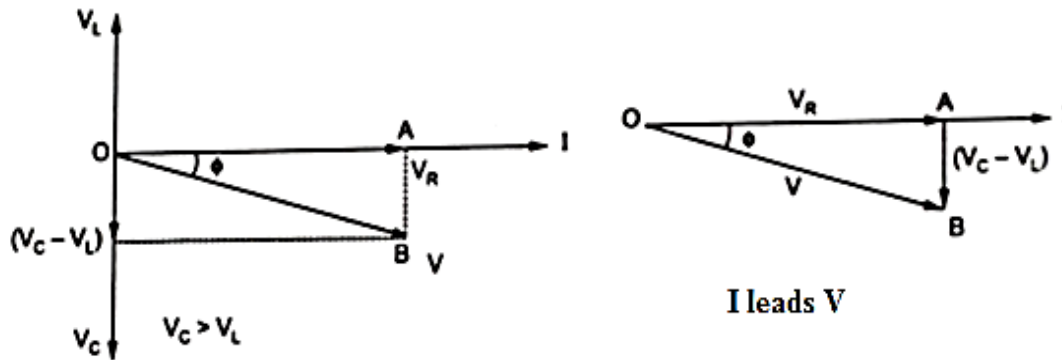
$$\text{where } Z = \sqrt{(R)^2 + (X_L - X_C)^2}$$

So, if $v = V_m \sin \omega t$, then $i = I_m \sin (\omega t - \phi)$ as current lags voltage by angle ϕ

$$X_L < X_C$$

When $X_L < X_C$, obviously, $I X_L$ i.e. V_L is less than $I X_C$ i.e. V_C . So, the resultant of V_L and V_C will be directed towards V_C . Current I will lead $(V_C - V_L)$.

The circuit is said to be capacitive in nature. The phasor sum of V_R and $(V_C - V_L)$ gives the resultant supply voltage V . This is shown in the Fig.



From the voltage triangle,

$$\begin{aligned} V &= \sqrt{(V_R)^2 + (V_C - V_L)^2} = \sqrt{(I R)^2 + (I X_C - I X_L)^2} \\ &= I \sqrt{(R)^2 + (X_C - X_L)^2} \end{aligned}$$

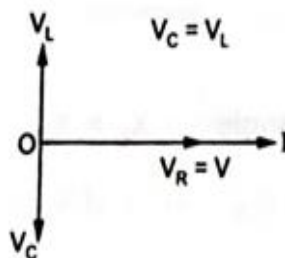
$$\therefore V = I Z$$

$$\text{where } Z = \sqrt{(R)^2 + (X_C - X_L)^2}$$

So, if $v = V_m \sin \omega t$, then $i = I_m \sin (\omega t + \phi)$ as current leads voltage by angle ϕ

$$X_L = X_C$$

When $X_L = X_C$, obviously, $V_L = V_C$. So, V_L and V_C will cancel each other and their resultant is zero. So, $V_R = V$ in such case and overall circuit is purely resistive in nature. The phasor diagram is shown in the Fig



From phasor diagram,

$$V = V_R$$

$$\therefore V = I R$$

$$\therefore V = I Z$$

$$\text{where } Z = R$$

Impedance

In general, for RLC series circuit impedance is given by,

∴

$$Z = R + j X$$

where

$$X = X_L - X_C = \text{Total reactance of circuit}$$

If $X_L > X_C$, X is positive and circuit is inductive.

If $X_L < X_C$, X is negative and circuit is capacitive.

If $X_L = X_C$, X is zero and circuit is purely resistive.

$$\tan \phi = \left[\frac{X_L - X_C}{R} \right], \quad \cos \phi = \frac{R}{Z} \quad \text{and} \quad Z = \sqrt{R^2 + (X_L - X_C)^2}$$

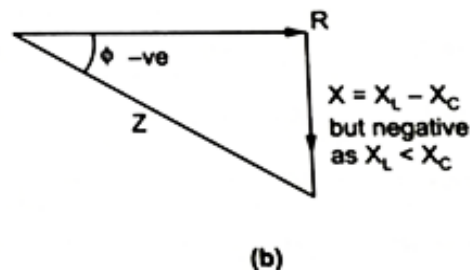
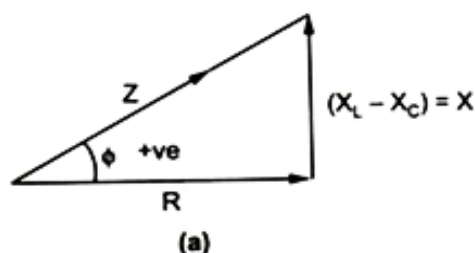
Impedance Triangle

The impedance is expressed as,

$$Z = R + j X \quad \text{where} \quad X = X_L - X_C$$

For $X_L > X_C$, ϕ is positive and the impedance triangle is as shown in the Fig. (a).

For $X_L < X_C$, $X_L - X_C$ is negative, so ϕ is negative and the impedance triangle is as shown in Fig.(b).



In both the cases, $R = Z \cos \phi$ and $X = Z \sin \phi$

Power and Power Triangle

The average power consumed by the circuit is,

$$P_{av} = \text{Average power consumed by } R + \text{Average power consumed by } L + \text{Average power consumed by } C$$

But, pure L and C never consume any power.

$$\therefore P_{av} = \text{Power taken by } R = I^2 R = I (I R) = I V_R$$

$$\text{But, } V_R = V \cos \phi \text{ in both the cases}$$

$$\therefore P = V I \cos \phi \text{ W}$$

Thus for any condition, $X_L > X_C$ or $X_L < X_C$, in general power can be expressed as

Power = Voltage X component of current in phase with voltage

$$P = V I \cos \Phi$$

A.C. Parallel Circuit

A parallel circuit is one in which two or more impedances are connected in parallel across the supply voltage. Each impedance may be a separate series circuit. Each impedance is called branch of the parallel circuit.

The Fig. shows a parallel circuit consisting of three impedances connected in parallel across an a.c. supply of V volts.

The current taken by each impedance is different.

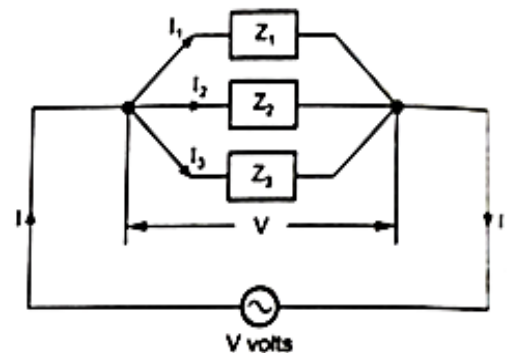


Fig. A.C. parallel circuit

Applying Kirchhoff's law,

$$\bar{I} = \bar{I}_1 + \bar{I}_2 + \bar{I}_3 \quad \dots \text{(Phasor addition)}$$

$$\therefore \frac{\bar{V}}{\bar{Z}} = \frac{\bar{V}}{\bar{Z}_1} + \frac{\bar{V}}{\bar{Z}_2} + \frac{\bar{V}}{\bar{Z}_3}$$

$$\therefore \frac{1}{\bar{Z}} = \frac{1}{\bar{Z}_1} + \frac{1}{\bar{Z}_2} + \frac{1}{\bar{Z}_3}$$

Where Z is called equivalent impedance. This result is applicable for 'n' such impedances connected in parallel.

Two Impedances in Parallel

If there are two impedances connected in parallel and if I_T is the total current, then current division rule can be applied to find individual branch currents.

$$\bar{I}_1 = \bar{I}_T \times \frac{\bar{Z}_2}{\bar{Z}_1 + \bar{Z}_2}$$

$$\bar{I}_2 = \bar{I}_T \times \frac{\bar{Z}_1}{\bar{Z}_1 + \bar{Z}_2}$$

Following are the steps to solve parallel a.c. circuit :

- 1) The currents in the individual branches are to be calculated by using the relation

$$\bar{I}_1 = \frac{\bar{V}}{\bar{Z}_1}, \quad \bar{I}_2 = \frac{\bar{V}}{\bar{Z}_2}, \quad \dots, \quad \bar{I}_n = \frac{\bar{V}}{\bar{Z}_n}$$

While the individual phase angles can be calculated by the relation,

$$\tan \phi_1 = \frac{X_1}{R_1}, \quad \tan \phi_2 = \frac{X_2}{R_2}, \quad \dots, \quad \tan \phi_n = \frac{X_n}{R_n}$$

- 2) Voltage must be taken as reference phasor as it is common to all branches.
- 3) Represent all the currents on the phasor diagram and add them graphically or mathematically by expressing them in rectangular form. This is the resultant current drawn from the supply.
- 4) The phase angle of resultant current I is power factor angle. Cosine of this angle is the power factor of the circuit..

Concept of Admittance

Admittance is defined as the reciprocal of the impedance. It is denoted by Y and is measured in unit siemens or mho.

Now, current equation for the circuit shown in the Fig. 4.32 is,

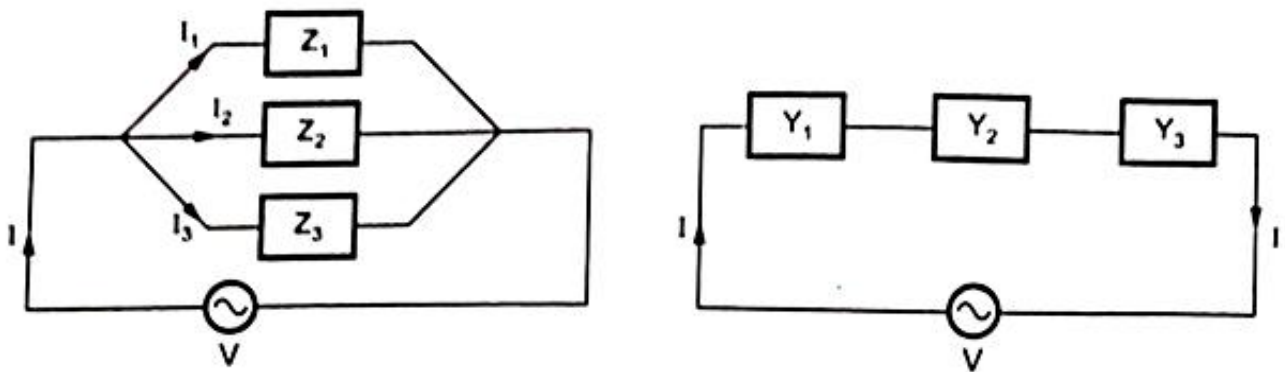
$$\bar{I} = \bar{I}_1 + \bar{I}_2 + \bar{I}_3$$

$$\bar{I} = \bar{V} \times \left(\frac{1}{Z_1} \right) + \bar{V} \times \left(\frac{1}{Z_2} \right) + \bar{V} \times \left(\frac{1}{Z_3} \right)$$

$$\bar{V}\bar{Y} = \bar{V}\bar{Y}_1 + \bar{V}\bar{Y}_2 + \bar{V}\bar{Y}_3$$

$$\therefore \bar{Y} = \bar{Y}_1 + \bar{Y}_2 + \bar{Y}_3$$

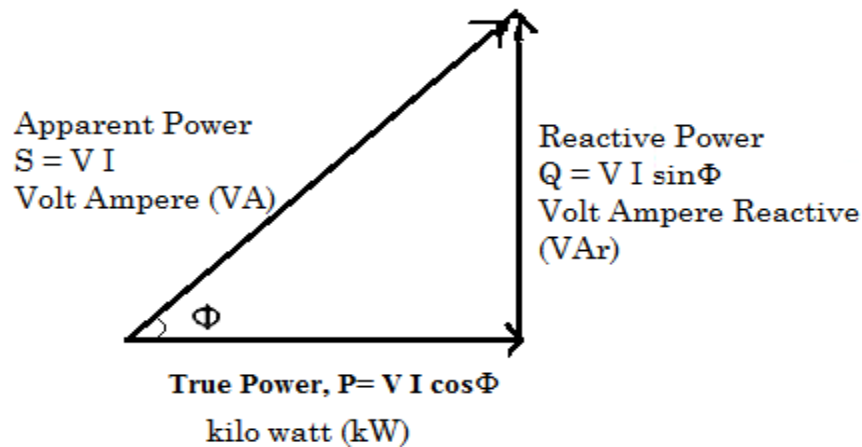
where Y is the admittance of the total circuit. The three impedances connected in parallel can be replaced by an equivalent circuit, where three admittances are connected in series, as shown in the Fig.



POWER: In an AC circuit, the various powers can be classified as

1. Real or Active power
2. Reactive power
3. Apparent power

Power Triangle: Power triangle is the representation of a right angle triangle showing the relation between active power, reactive power and apparent power. When each component of the current that is the active component ($I\cos\phi$) or the reactive component ($I\sin\phi$) is multiplied by the voltage V , a power triangle is obtained shown in the figure below



True power/Real power/Active power: The power due to the active component of current is called as the active power or real power. It is the power that is actually consumed or utilized in an AC Circuit and does useful power. It is denoted by 'P' and measured in kilowatt (kW) or MW.

$$P = V * I \cos \Phi$$

$$= I^2 R \cos \Phi$$

Where V= rms value of voltage
 I= rms value of current
 $\cos \Phi$ = power factor

Reactive power: The power due to the reactive component of current is called as the reactive power. It is denoted by Q. Reactive power does not do any useful work. It is the circulating power in the L and C components. The unit for reactive power is Volt Amperes Reactive (VAr).

$$Q = V * I \sin \Phi$$

$$= I^2 X_L \sin \Phi$$

Apparent power: The apparent power is the total power in the circuit. It is the product of root mean square (RMS) value of voltage and current. It is denoted by 'S'. The unit for apparent power is Volt Amperes (VA) and measured in KVA or MVA.

$$S = V I$$

$$= I^2 Z$$

$$S^2 = P^2 + Q^2$$

Power factor: It is the cosine of the angle between V and I. From the power triangle shown above the power, the factor may be determined by taking the ratio of true power to the apparent power.

$$\text{Power factor, } \cos\Phi = \frac{\text{Real Power}}{\text{Apparent Power}} = \frac{\text{kW}}{\text{kVA}}$$

$$\text{Or, } \cos\Phi = \frac{\text{Resistance}}{\text{Impedance}} = \frac{R}{Z}$$

Importance of Power Factor: A power factor of one or "unity power factor" is the goal of any electric utility company since if the power factor is less than one, they have to supply more current to the user for a given amount of power use. In so doing, they incur more line losses. They also must have larger capacity equipment in place than would be otherwise necessary. As a result, an industrial facility will be charged a penalty if its power factor is much different from 1.

Causes of Low power factor: The reason for the low power factor is because of the inductive load. The current in the inductive load lag behind the voltage. The power factor is therefore lagging. The important inductive loads responsible for the low power factor are the three-phase induction motors (which operate at a 0.8 lagging power factor), transformer, lamps and welding equipment operate at low lagging power factors. Power factor improvement methods are used for improving the value of power factor in a power system.

Disadvantages of low power factor

- Higher current is required by the equipment, due to which the economic cost of the equipment is increased.
- At low power factor, the current is high which gives rise to high copper losses in the system and therefore the efficiency of the system is reduced.
- Higher current produced a large voltage drop in the apparatus. This results in the poor voltage regulation.

Power Factor Improvement: If the power factor is low or poor, it is necessary to improve or correct it. It may be improved by injecting a leading current into the circuit so as to neutralize the effect of lagging current. The power factor may be improved by using static capacitors or synchronous motors.

Three Phase A.C Circuits:

Advantages of three phase system: In the three phase system, the alternator armature has three windings and it produces three independent alternating voltages. The magnitude and frequency of all of them are equal but they have a phase difference of 120° between each other. Such a three phase system has following advantages over single phase system:

1. The output of three phase machine is always greater than single phase machine of same size, approximately 1.5 times. So for a given size and voltage a three phase alternator occupies less space and costs less than single phase having same rating.
2. For transmission and distribution, three phase system needs less copper or less conducting material than single phase system for given volt amperes and voltage rating, so transmission becomes very much economical.
3. It is possible to produce rotating magnetic field with stationary coils by using three phase system. Hence three phase motors are self-starting.
4. In single phase system, the instantaneous power is a function of time and hence fluctuates with respect to time. This fluctuating power causes considerable vibrations in single phase motors. Hence performance of single phase motors is poor. While instantaneous power in symmetrical three phase system is constant.
5. Single phase supply can be obtained from three phase but three phase cannot be obtained from single phase.
6. Power factor of single phase motor is poor than three phase motors of same rating.

But it is found that optimum number of phases required to get all above said advantages is three. Any further increase in number of phases cause a lot of complications. Hence three phase system is accepted as standard poly-phase system throughout the world.

Generation of 3-phase E.M.F.

In the 3-phase system, there are three equal voltages of the same frequency but displaced from one another by 120° electrical. These voltages are produced by a three-phase generator which has three identical windings or phases displaced 120° electrical apart. When these windings are rotated in a magnetic field, e.m.f. is induced in each winding or phase. These e.m.f.s are of the same magnitude and frequency but are displaced from one another by 120° electrical.

Consider three identical coils a_1a_2 , b_1b_2 and c_1c_2 as shown in the above figure. In this figure a_1 , b_1 and c_1 are the starting terminals, whereas a_2 , b_2 and c_2 are the finish terminals of the three coils. The phase difference of 120° has to be maintained between the starts terminals a_1 , b_1 and c_1 . Now, let the three coils mounted on the same axis, and they are rotated by either keeping coil stationary and moving the magnetic field or vice versa in an anticlockwise direction at (ω) radians per seconds. Three e.m.f.s are induced in the three coils respectively.

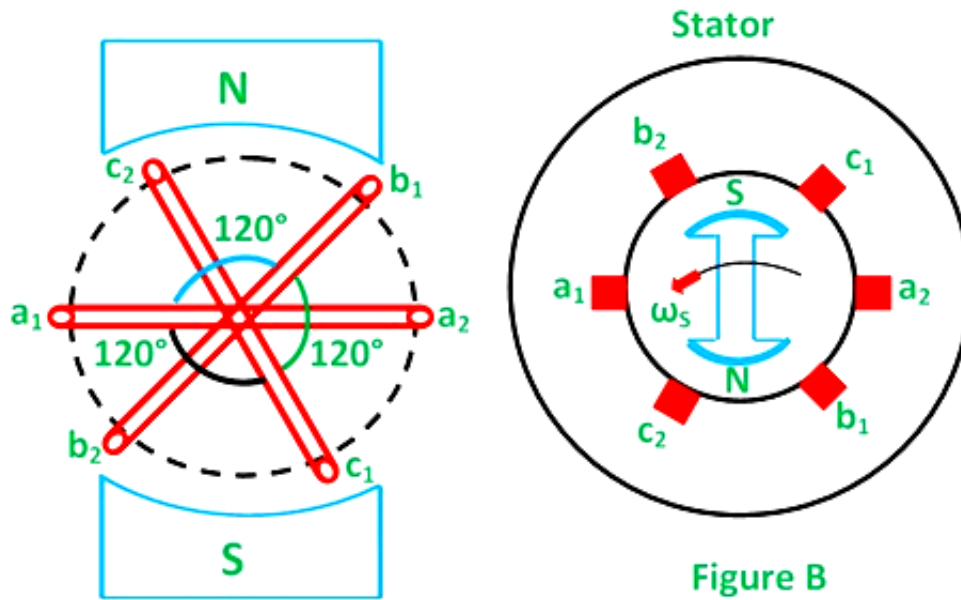


Figure A

Figure A: Three-phase e.m.f generation

When the coil a_1a_2 is in the position AB shown in the above figure, the magnitude and direction of the e.m.f.s induced in the various coils is as under:

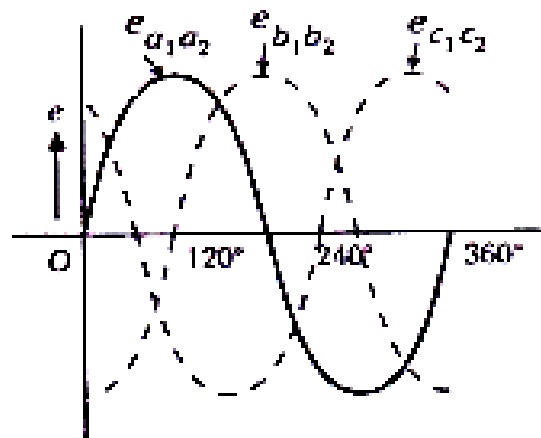


Figure B: Three-phase Sinusoidal waveform

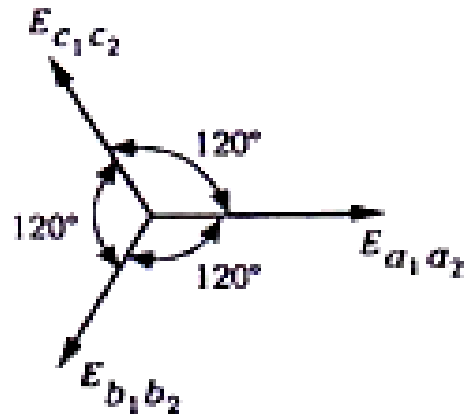
The e.m.f. induced in the coil a_1a_2 is zero and is increasing in the positive direction as shown by the waveform represented as $e_{a_1a_2}$.

The coil b_1b_2 is 120° electrically behind the coil a_1a_2 . The e.m.f. induced in this coil is negative and is becoming maximum negative as shown by the wave $e_{b_1b_2}$.

Similarly, the coil c_1c_2 is 120° electrically behind the coil b_1b_2 , or we can also say that the coil c_1c_2 is 240° behind the coil a_1a_2 . The e.m.f. induced in the coil is positive and is decreasing as represented by the waveform $e_{c_1c_2}$.

Phasor Diagram

The e.m.f.s induced in the three coils in a 3 phase circuits are of the same magnitude and frequency and are displaced by an angle of 120° from each other as shown below in the phasor diagram.



These EMFs of a 3 phase circuits can be expressed in the form of the various equations given below.

$$e_{a_1a_2} = E_m \sin \omega t$$

$$e_{b_1b_2} = E_m \sin(\omega t - 2\pi/3) = E_m \sin(\omega t - 120^\circ)$$

$$e_{c_1c_2} = E_m \sin(\omega t - 4\pi/3) = E_m \sin(\omega t - 240^\circ)$$

Phase sequence: The order in which the voltages in the three phase system reach their maximum positive values is called the phase sequence.

For example, in Fig. (A) the three coils a_1a_2 , b_1b_2 and c_1c_2 , are rotating in anticlockwise direction in the magnetic field. The coil a_1a_2 is 120° electrical ahead of coil b_1b_2 and 240° electrical ahead of coil c_1c_2 .

Therefore, e.m.f. in coil a_1a_2 leads the e.m.f. in coil b_1b_2 by 120° and that in coil c_1c_2 by 240° . It is evident from Fig (B) that $e_{a_1a_2}$ attains maximum positive first, then $e_{b_1b_2}$ and $e_{c_1c_2}$. Hence, the phase sequence is a,b,c. The 3 phases may be numbered (1, 2, 3) or lettered (a, b, c) or specified by colors (R Y B). By normal convention, sequence RYB is considered positive and R B Y negative.

It is necessary to employ some systematic notation for the solution of a.c. circuits and systems containing a number of e.m.f.s acting and currents flowing so that the process of solution is simplified and less prone to errors. It is normally preferred to employ double-subscript notation

while dealing with a.c. electrical circuits. In this system, the order in which the subscripts are written indicates the direction in which e.m.f. acts or current flows.

For example, if e.m.f. is expressed as E_{ab} , it indicates that e.m.f. acts from a to b; if it is expressed as E_{ba} , then the e.m.f. acts in a direction opposite to that in which E_{ab} acts.

i.e., $E_{ba} = -E_{ab}$

Similarly, I_{ab} indicates that current flows in the direction from a to b but I_{ba} indicates that current flows in the direction from b to a;

i.e., $I_{ba} = -I_{ab}$.

Balanced Supply and Load

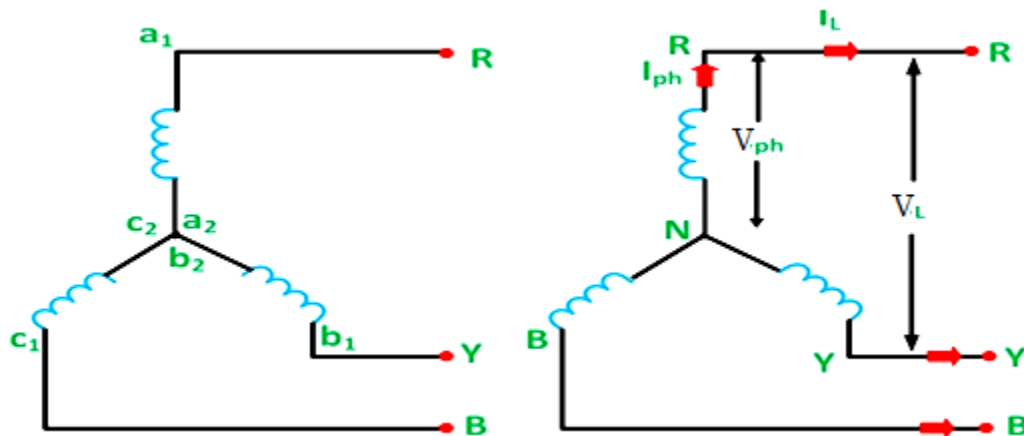
Balanced Supply: A supply is said to be balanced when the three phase voltages are equal, and the phase difference is 120° between one another.

Balanced load: A load is said to be balanced if the impedance of the three phases or three circuit loads are equal.

When a balanced supply, where the three phase voltages are equal, and the phase difference is 120° between one another, supplies balanced load, where the impedance of the three phases or three circuit loads are equal, then the current flowing through these three phases will also be equal in magnitude, and will also have a phase difference of 120° with one another. Such an arrangement is called a balanced load.

Star Connection in a 3 Phase System

In the **Star Connection**, the similar ends (either start or finish) of the three windings are connected to a common point called star or neutral point. The other three conductors run from the remaining three free terminals called **line conductors**. The wires are carried to the external circuit, giving three phase, three wire star connected systems. However, sometimes a fourth wire is carried from the star point to the external circuit, called **neutral wire**, forming three phase, four wire star connected systems. The star connection is shown in the diagram below.



The finish terminals a_2 , b_2 , and c_2 of the three windings are connected to form a star or neutral point. The three conductors named as R, Y and B run from the remaining three free terminals as shown in the above figure.

The current flowing through each phase is called Phase current I_{ph} , and the current flowing through each line conductor is called Line Current I_L . Similarly, the voltage across each phase is called Phase Voltage V_{ph} , and the voltage across two line conductors is known as the Line Voltage V_L .

Relation between Phase Voltage and Line Voltage in Star Connection

1. Line voltage is root 3 times of phase voltage.

i.e, Line voltage = $\sqrt{3}$ Phase voltage

$$V_L = \sqrt{3} V_{ph}$$

2. Line Current is equal to Phase Current.

i.e, Line current = Phase current

$$I_L = I_{ph}$$

3. Power output per phase = $V_{ph} I_{ph} \cos \Phi$

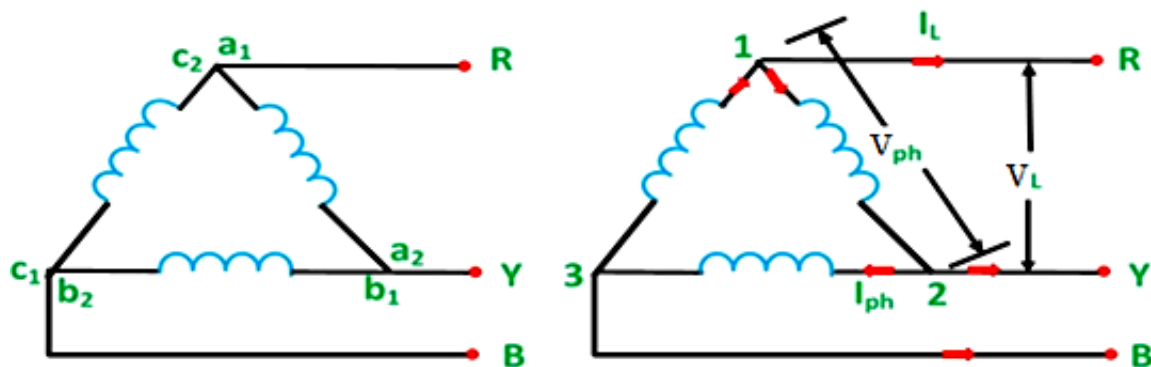
Total power output, $P = 3 V_{ph} I_{ph} \cos \Phi$

or, Power = $\sqrt{3} V_L I_L \cos \Phi$

Where, $\cos \Phi$ is the power factor of the load

Delta Connection in a 3 Phase System

In **Delta (Δ) or Mesh connection**, the finished terminal of one winding is connected to start terminal of the other phase and so on which gives a closed circuit. The three conductors running from the three junctions of the mesh are called Line Conductors. The connection in Delta form is shown in the figure below.



To obtain the Delta connection, a_2 is connected with b_1 , b_2 is connected with c_1 and c_2 is connected with a_1 , as shown in the above figure. The three conductors R, Y and B are running from the three junctions known as Line Conductors. The current flowing through each phase is called Phase Current

(I_{ph}), and the current flowing through each line conductor is called Line Current (I_L). The voltage across each

phase is called Phase Voltage (V_{ph}), and the voltage across two line conductors is called Line Voltage (V_L).

Relation between Phase Voltage and Line Voltage in Delta Connection

1. Line Voltage is equal to Phase Voltage
i.e, $V_L = V_{ph}$
2. Line current is root three times of phase current.
Line current $= \sqrt{3}$ Phase current
 $I_L = \sqrt{3} I_{ph}$
3. Power output per phase $= V_{ph} I_{ph} \cos \Phi$
Total power output, $P = 3 V_{ph} I_{ph} \cos \Phi$
or, Power $= \sqrt{3} V_L I_L \cos \Phi$
Where, $\cos \Phi$ is the power factor of the load